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Definition of catchment area in karst: case of the rivers Krčić and Krka, Croatia

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Abstract The study area consists of the spring zones of the Krčić, Krka and Cetina river catchments located in the Dinaric karst, Croatia. Classical hydrological approaches and some newer time and frequency domain methods are used in order to validate the existing hypotheses both qualitatively and quantitatively, and these contribute to factual information about the hydrological behaviour of the catchments. The groundwater recharge rates are calculated by a mathematical model based on Palmer's soil-moisture balance method. The values of parameters of the groundwater recharge model are estimated by the spectral method. The calculated monthly and annual groundwater recharge rates form the basis for estimating the hydrological catchment areas of the spring zones and also for the determination of quantitative relationships between the catchments.

Key words catchment area; Croatia; effective rainfall; groundwater recharge; karst hydrology; karst spring

Définition de l'aire d'un bassin versant karstique: cas des rivières Krčić et Krka (Croatie)

Résumé La zone d'étude correspond aux zones de sources des bassins versants des rivières Krčić, Krka et Cetina dans le karst des Dinarides. Les méthodes hydrologiques classiques, ainsi que certaines nouvelles méthodes dans les domaines temporel et fréquentiel, sont utilisées afin de valider qualitativement et quantitativement les hypothèses existantes, et de contribuer à la caractérisation factuelle du fonctionnement de ces bassins versants. Les taux de recharge hydrogéologique sont calculés au moyen d'un modèle mathématique basé sur la méthode du bilan de l'humidité du sol de Palmer. Les valeurs des paramètres du modèle de recharge hydrogéologique sont estimées par méthode spectrale. Les taux de recharge mensuels et annuels calculés sont à la base de l'estimation des aires des bassins versants des zones de sources, ainsi que de la détermination de relations quantitatives entre les bassins.

Mots clefs aire de bassin versant; Croatie; pluie efficace; recharge hydrogéologique; hydrologie karstique; source karstique

INTRODUCTION

The determination of catchment boundaries and catchment area is the starting point in all hydrological analyses and essential as the basis for all hydrological calculations (Bonacci, 1987). It is always a very complex task, especially in karst. Groundwater exchanges with adjacent aquifers through underground routes or inflows from surface streams and accumulations are common in karst. Consequently, it is only in exceptional cases that surface (topographic or orographic) and subsurface (hydrological or hydrogeological) catchment boundaries coincide. The problem is further complicated by the time-variant hydrological boundaries that are dependent on fluctuations in groundwater levels.

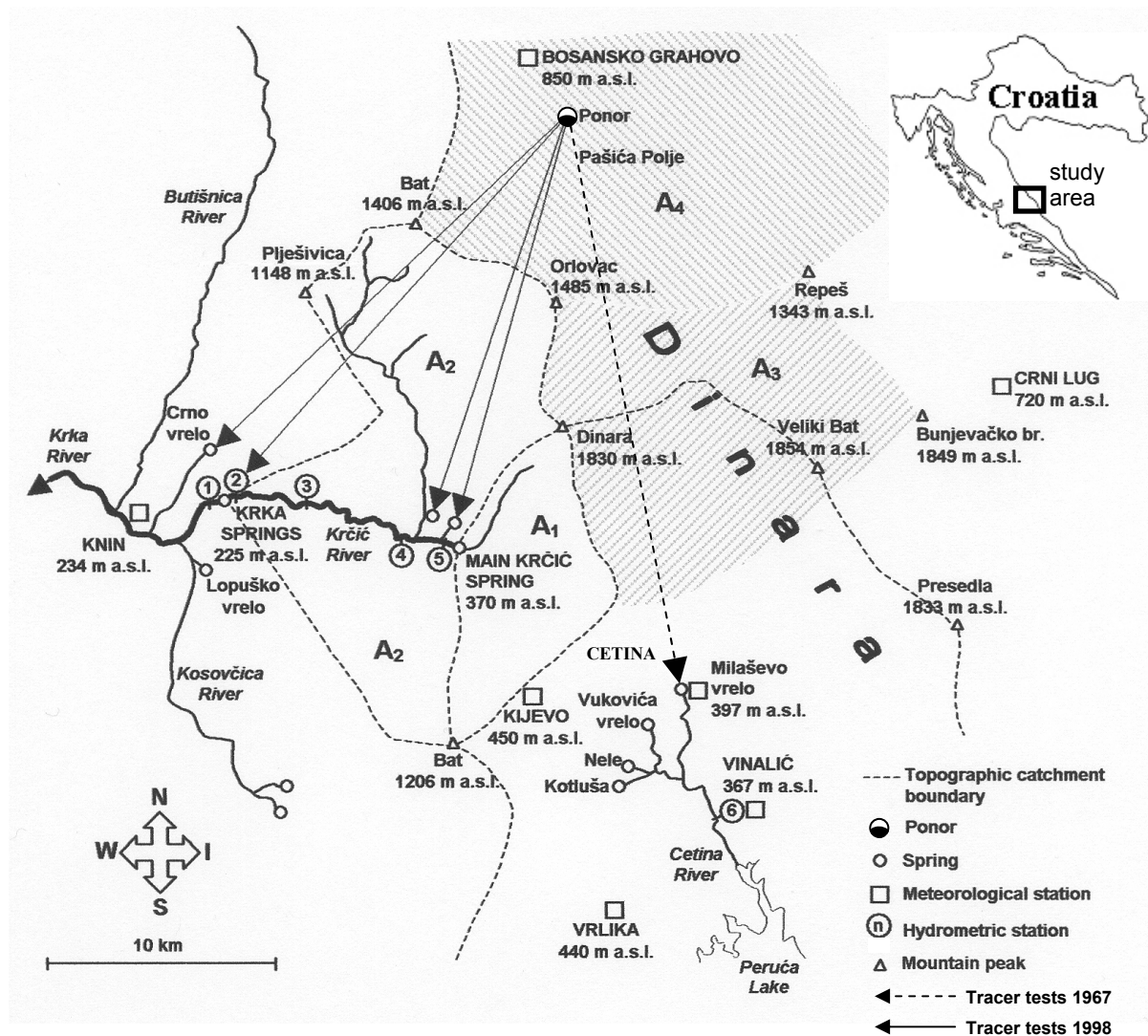
In order to define the hydrological catchment boundaries more precisely, it is necessary to carry out extensive geological, hydrological and hydrogeological investigations. According to Perrin *et al.* (2000), delimitations based on geological considerations cannot be completely reliable, because they seem to overestimate the hydrological catchment area. Hydrogeological investigations involve determinations of connectivity between different locations using various tracing methods, which are

usually too expensive to carry out. Consequently, where geological and hydrogeological investigations are limited, the estimation of hydrological catchment area (recharge area) and the delimitation of catchment boundaries are oriented toward hydrological analyses. If the number and the locations of precipitation stations are convenient, linear transfer functions can be applied (Dreiss, 1983), but this method is not recommendable for karst springs with dominant diffuse flow. Bonacci (1987) used the groundwater hydrograph method to estimate the hydrological catchment area of the Jadro karst spring, Croatia. This procedure involves the calculation of spring hydrographs by a mathematical expression and the minimization of differences between the calculated and observed hydrographs by varying the value of the hydrological catchment area. The mathematical expression is derived by assuming that the system unit response function has form of the Zoch model (Singh, 1988). Instead of the Zoch model, Denić-Jukić & Jukić (2003) used the composite transfer functions in a similar way. For the purpose of preliminary investigations, the hydrological catchment area can be estimated with various empirical expressions for runoff deficit calculations and regional formulae (Bonacci, 1987, 1999; Bonacci & Magdalenić, 1993), or on the basis of regional precipitation and evapotranspiration maps (Petrič, 2002).

Regardless of the method applied, the prerequisite for estimating hydrological catchment area is the determination of effective rainfall or groundwater recharge by using very simple regional formulae and empirical expressions, or more complicated mathematical models. In addition to the estimation of hydrological catchment area, this study shows that monthly and annual groundwater recharge rates calculated by the mathematical model can also be valuable in preliminary analyses for the investigation of the hydrological behaviour of catchments. This is especially important in karst, where catchments have time-variant hydrological boundaries, i.e. unique values of hydrological catchment areas frequently do not exist.

STUDY AREA, AVAILABLE DATA AND RESULTS OF BASIC HYDROLOGICAL ANALYSES

The study area consists of the springs zone in the catchments of the karst rivers Krčić, Krka and Cetina, located in the bare Dinaric karst in Croatia (Fig. 1). The Main Krčić Spring is situated near the base of the Dinara Mountains at an altitude of 370 m a.s.l. Several small springs are located in the vicinity of this spring, near the right bank of the River Krčić. After approximately 10 km, the River Krčić ends with a waterfall in the River Krka. The most important of the Krka springs (Main Spring) is located in the cave directly beneath the waterfall at 225 m a.s.l. The Krka springs zone also contains two small springs: Small Spring and Third Spring (Bonacci, 1985, 1987). The contribution of the two small springs to the total Krka springs discharge is approximately 10–20%. The springs zone of the Cetina River has four springs: Milaševo vrelo, Vukovića vrelo, Nele and Kotluša (Fig. 1). They are about 12 km away from the Main Krčić Spring. Tracer tests performed in 1998 of the ponor in the Pašića Polje (Fig. 1) revealed direct groundwater connections between the Pašića Polje and both the Krka springs and the small springs located near the right bank of the River Krčić (Fig. 1). No connection was found with either the Main Krčić Spring or the Cetina springs. The results of these 1998 tracer tests are completely different from those performed in 1967 when only the connection with the Cetina springs was established.



A₁ – topographic catchment area of the Main Krčić Spring.

A₁+A₂ – topographic catchment area of the Krčić River and Krka springs.

A₃ – hydrological catchment area of the Main Krčić Spring and Cetina springs.

A₄ – hydrological catchment area of the Krka springs, Krčić River and probably Cetina springs.

Fig. 1 Location map of the Main Krčić Spring, Krka and Cetina springs with topographic catchment boundaries, positions of meteorological and hydrometric gauging stations and results of tracer tests from the ponor in the Pašića Polje.

The Krčić catchment is composed of Upper Triassic dolomites, Jurassic dolomites and limestone and Quaternary alluvial deposits. The River Krčić flows partly through Upper Triassic dolomites and partly through the limestone overlying the dolomites. The Krka and Cetina springs catchment zones are formed mainly of Jurassic limestone. The Dinaric karst is deep and well developed. It is practically without soil cover, with scarce vegetation and with numerous surface and subsurface micro and macro karst phenomena (Bonacci, 1987).

The Cetina springs discharge has been measured daily since 1946 at the Cetina-Vinalić station (Station 6 on Fig. 1). Discharges of the Krka and Krčić springs are not well documented. Reliable daily data, necessary for analyses, are available only for the period 1982–1990. Five gauging stations were located in the area of the Krčić and

Table 1 Basic information about the gauging stations along the Krka and Krčić rivers.

Station no. (Fig. 1)	River	Gauging station	Datum plane (m a.s.l.)	Distance from the R. Krčić mouth (km)	Period of measurement
1	Krka	Topolje	214.86	-1.00	1950–1990
2	Krčić	Waterfall	261.32	0.02	1979–1983
3	Krčić	Krčić	272.85	2.55	1950–1966; 1975–1990
4	Krčić	Pregrada	338.16	8.23	1979–1990
5	Krčić	Spring	364.74	9.91	1980–1984

Krka springs before 1990: Krka-Topolje, Krčić-Waterfall, Krčić-Krčić, Krčić-Pregrada and Krčić-Spring (stations 1, 2, 3, 4 and 5, respectively, on Fig. 1). Basic information about the gauging stations is presented in Table 1. The Krčić-Waterfall and Krčić-Spring stations were active only during the periods 1979–1983 and 1980–1984, respectively. Consequently, the total Krka springs discharge is estimated by subtracting discharge observed at Krčić-Krčić station from that observed at Krka-Topolje, whereas the discharge from Krčić-Pregrada station represents the Krčić springs discharge. The average annual discharge values for the Krčić, Krka and Cetina springs are 3.93, 6.54 and 10.40 m³ s⁻¹, respectively.

Available meteorological data for the period 1982–1990 are: daily rainfall rates, temperatures and relative humidity from the meteorological stations Knin and Kijevo (Fig. 1). Annual rainfall rates from the meteorological stations: Kijevo, Knin, Vrlika, Vinalić, Cetina springs, Crni Lug and Bosansko Grahovo (Fig. 1) for the periods 1964–1971 and 1981–1985 are also used. The lowest average annual rainfall rate is observed at the Knin meteorological station, which is located at the lowest altitude (234 m a.s.l.). The highest annual rainfall rates are measured at Bosansko Grahovo meteorological station, located at the highest altitude (850 m a.s.l.). The average annual rainfall over the study area, estimated by using available data, is 1550 mm for the period 1982–1990 and 1360 mm for 1964–1971. The average annual temperature at the Knin meteorological station is 13°C, while the annual average relative humidity is 64%. The climate is continental influenced by the mild Mediterranean climate. The summers are hot, with the maximum temperature above 30°C, whilst the winters are cold with temperatures as low as -10°C. The average annual rainfall over the catchment and the total annual discharges from the Krčić, Krka and Cetina springs for the period 1982–1990 are presented in Table 2.

The Cetina and Krka springs are perennial, with minimum discharges greater than 1.0 and 2.0 m³ s⁻¹, respectively, while the Krčić springs dry up every year. The relationships between average monthly and annual discharges of the Krka springs and discharge at the Krka-Topolje station are presented in Fig. 2. The area between ordinate axes and the line $y = x$ represents the discharge flowing into the River Krka exclusively from under ground, whereas the second half of the quadrant represents the discharge supplied by surface flow from the River Krčić. The points on the line $y = x$ represent the situations when there was no flow in the River Krčić. From Fig. 2, it can be seen that the River Krčić was dry when the Krka springs discharge was less than 6.31 m³ s⁻¹, but in most cases, it dried up when the Krka springs discharge dropped below 4.20 m³ s⁻¹. The average and maximum annual discharges observed along the River Krčić are shown in Table 3. The average annual discharge at the Krčić-Krčić station is usually smaller than that at the Krčić-Pregrada and Krčić-Spring stations. It

Table 2 The annual rainfall rates and the total annual discharges from the Krčić, Krka and Cetina springs for eight hydrological years.

Hydrological year	P_A (m)	V_A (10^6 m ³):		
		Krčić springs	Krka springs	Cetina springs
1982/83	1.620	148.36	162.63	372.47
1983/84	1.742	117.98	235.49	381.43
1984/85	1.575	130.50	206.19	355.94
1985/86	1.906	189.92	231.53	420.04
1986/87	1.287	114.27	185.68	312.52
1987/88	1.853	128.28	256.05	335.39
1988/89	1.074	73.85	181.29	198.35
1989/90	1.068	71.77	191.56	217.28
Σ	12.125	974.93	1650.41	2593.42

P_A : annual rainfall rate; V_A : total annual discharge.

Table 3 Characteristic annual discharges (in m³ s⁻¹) measured at four gauging stations along the River Krčić.

No.	Year	Average:				Maximum:			
		Spring	Pregrada	Krčić	Waterfall	Spring	Pregrada	Krčić	Waterfall
1.	1975			4.72				67.8	
2.	1976			5.43				31.3	
3.	1977			5.67				24.8	
4.	1978			6.24				35.0	
5.	1979		7.19	6.01			27.4	32.5	
6.	1980	8.38	7.68	6.23	6.32	20.7	27.7	46.9	54.4
7.	1981	5.74	5.25	4.56	5.18	18.9	31.1	52.1	65.4
8.	1982	5.57	4.98	4.01	5.11	22.5	34.7	49.2	57.7
9.	1983	3.54	3.07	3.03	3.95	13.4	18.8	21.2	32.0
10.	1984	4.98	4.86	4.98		11.7	21.5	24.7	
11.	1985		3.65	2.64			21.6	15.9	
12.	1986		5.45	3.89			26.2	24.3	
13.	1987		3.87	3.54			22.6	25.7	
14.	1988		3.87	3.39			17.4	16.4	
15.	1989		2.84	2.80			16.3	15.7	
16.	1990		2.97	2.38			22.4	27.0	

shows a strong influence of river-bed karstification on the River Krčić flow, which causes significant water losses along the river bed. In Fig. 3, a longitudinal cross-section of the Krčić river bed with the minimum and maximum groundwater levels is shown, together with the position of the zone of strong karstification. Actually, the flow along the River Krčić occurs partly on the surface and partly as subsurface flow (Bonacci, 1985, 1987). The hypothetical karst pipe, according to Hajdin & Ivetić (1978), is also shown in Fig. 3. The assumptions of the existence of a pipe system, and of the dimensions of the pipe, are speculative, based on a conceptual approach describing the intensive subsurface flow to the Krka springs through a highly karstified zone.

There is no reliable recognition of hydrological catchment areas for the Krčić and Krka springs. The hydrological catchment areas of these springs and the Krka-Topolje station were estimated for the period 1982–1990 as the ratio between total annual

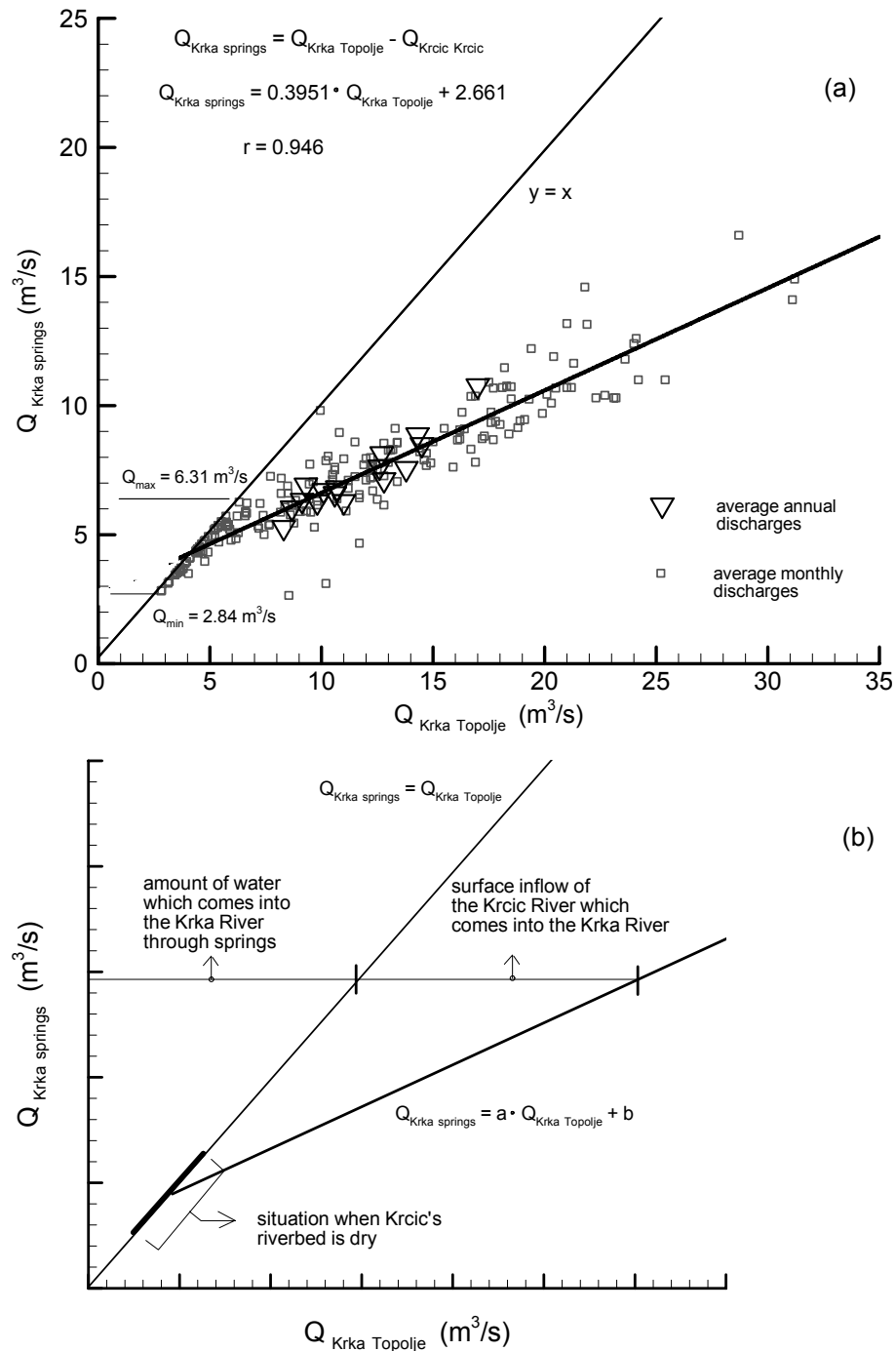


Fig. 2 (a) Relationship between average monthly and annual discharges of the Krka springs and Krka-Topolje for the period 1975–1990 and (b) schematic explanation of this relationship.

spring discharges and effective annual rainfall, as displayed in Table 4. Effective rainfall was calculated by using the regional formula of Srebrenović (1986) and the empirical expressions of Turc (1954) and Coutagne (1954), which have already been applied in preliminary hydrological analyses on several catchments in the Dinaric karst (Bonacci, 1987, 1993, 1999, 2001; Bonacci & Magdalenic, 1993) and other karst areas

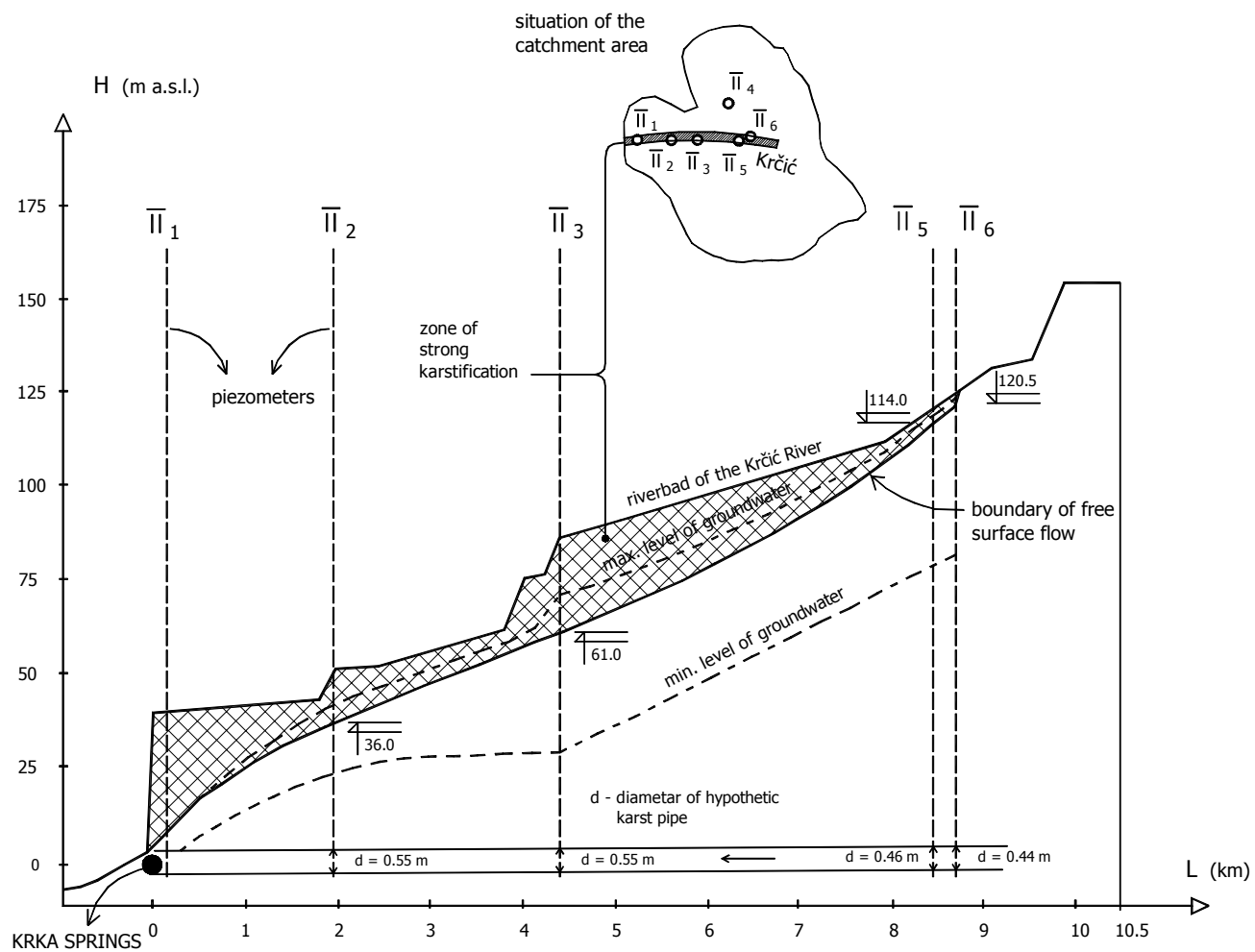


Fig. 3 Longitudinal cross-section of the Krčić river bed (Bonacci, 1985).

Table 4 Hydrological catchment areas calculated by regional formulas of Srebrenović (1986) and the empirical expressions of Turc (1954) and Coutagne (1954) for the period 1982–1990.

Station	Hydrological catchment area (km ²) calculation:		
	Turc [†]	Coutagne [‡]	Srebrenović*
Main Krčić spring	137	148	118
Krka springs	232	252	199
Krka - Topolje	349	379	300

* $P_E = 0.88P + 250f - 494$.

[†] $D = P(0.9 + (P^2/L^2))^{-0.5}$; $L = 300 + 25T + 0.05T^2$.

[‡] $D = P - \lambda P^2$ for $\frac{1}{8}\lambda \leq P < \frac{1}{2}\lambda$; $D = P$ for $P < \frac{1}{8}\lambda$; $D = 0.20 + 0.035T$ for $P \geq \frac{1}{2}\lambda$;

$$\lambda = \frac{1}{0.8 + 0.14T}.$$

P : total annual rainfall; P_E : effective annual rainfall; f : karst factor; T : average annual temperature;

D : runoff deficit ($= P - P_E$).

(Soulis, 1984). The average hydrological catchment areas obtained by the three methods are: 228 km² for the Krka springs, 134 km² for the Main Krčić Spring and 343 km² for the Krka-Topolje station. Since the topographic catchment areas of the Main Krčić Spring and the Krka springs are only 35 km² (area A_1 in Fig. 1) and 145 km² (area $A_1 + A_2$ in Fig. 1), respectively, it is evident that significant parts of both catchments are located outside the topographic catchment boundaries.

Recent investigations (Jukić, 2005a,b) revealed that the Main Krka Spring is an ascending karst spring whose aquifer is located deep inside the karst massif, whereas the Krčić springs function as descending karst springs. Following these results, as well as those of previous studies (Bonacci, 1985, 1987) and tracer tests performed in 1998, Bonacci & Ljubenković (2005) made a schematic representation of the hydrological behaviour of the Krčić and Krka springs catchments, as presented in Fig. 4. From the higher area A_4 , water flows through the ponors in the Pašića Polje to the River Krčić by underground percolation (Q_3'). Surface discharge of the River Krčić (Q_1) includes water from the Krčić springs (Q_{IK}) and also water that enters the river from the topographic catchment area between the Krka springs and the waterfall. Part of the River Krčić surface flow seeps into the karst under ground and enters the Krka spring (Q_2). This flow is referred to as subsurface flow (Bonacci, 1985, 1987), to distinguish it from the deep underground water (Q_3'') flowing to the Krka spring from higher areas (A_4). However, the connections between the Pašića Polje and the Cetina springs, established in 1967, as well as the origin of water at the Main Krčić Spring have not been explained completely yet, because the tracer tests performed in 1998 were not able to confirm such a connection.

Although the topographic catchment of the Main Krčić Spring is located completely within that of the Krka springs (Fig. 1), recent investigations (Jukić, 2005b) have shown that the hydrological connections between the Krčić and Cetina springs are stronger than those between the Krčić and Krka springs. The shapes of flow duration curves plotted on lognormal probability paper are also in accordance with this hypothesis (Fig. 5). Soulis (1985) analysed the flow duration curves of several karst springs in Greece. The breaks in the flow duration curves plotted on normal or lognormal probability paper were ascribed to the spilling-over of the karst aquifer

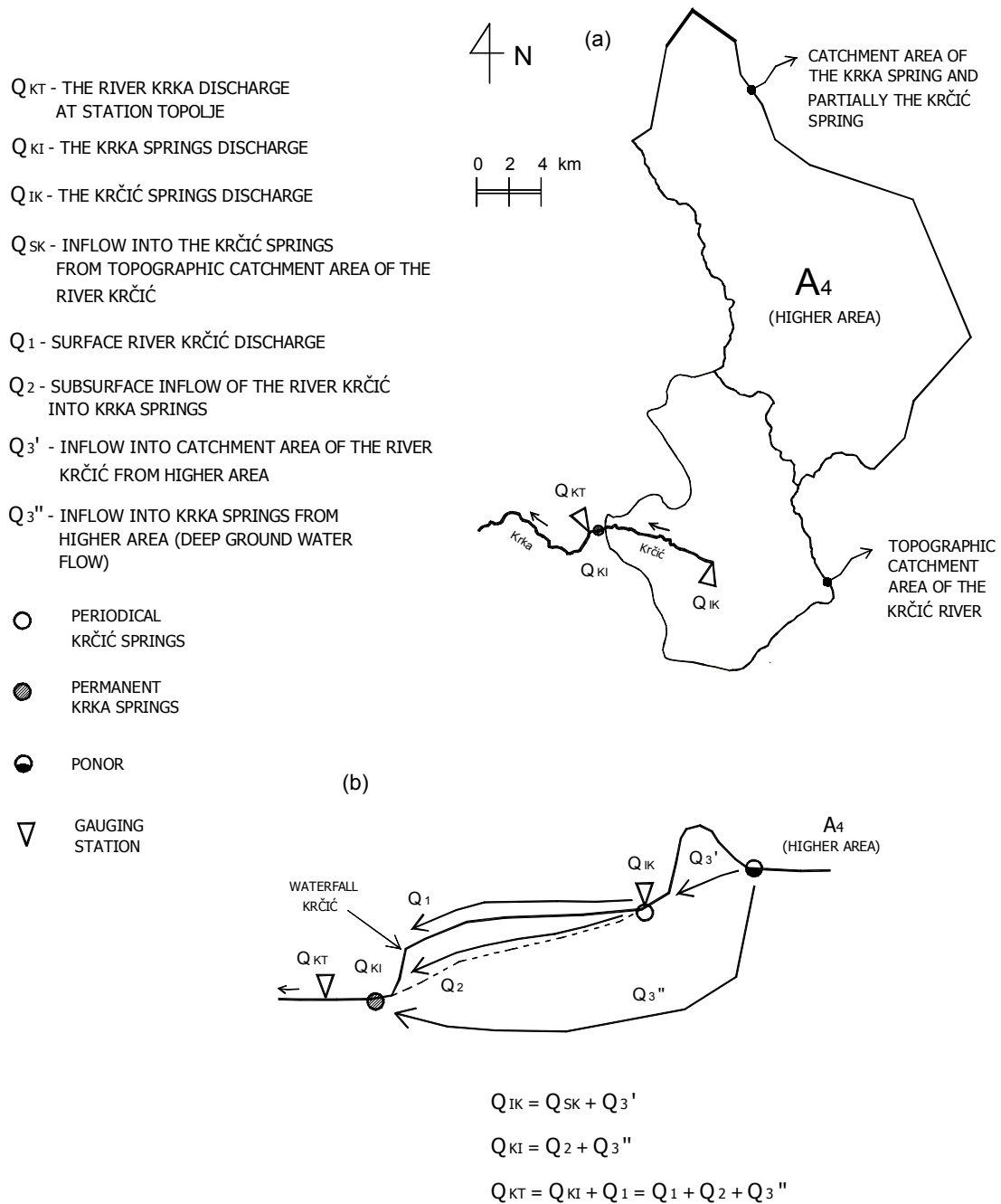


Fig. 4 (a) Location map and (b) schematic representation of longitudinal cross-section showing the hydrological behaviour of the springs in the Krčić and Krka river catchments (Bonacci & Ljubenkov, 2005).

towards external outlets or catchments and to the recharging by waters from surrounding areas. The break in the curve for the Krka-Topolje station, located at the average daily discharge of about $6 \text{ m}^3 \text{ s}^{-1}$, is evident and coincides with the beginning of intensive surface flow along the River Krčić (Fig. 2). The flow duration curves indicate that the flow of the River Krčić represents inflows from the standpoint of the River Krka catchment at the Krka-Topolje station.

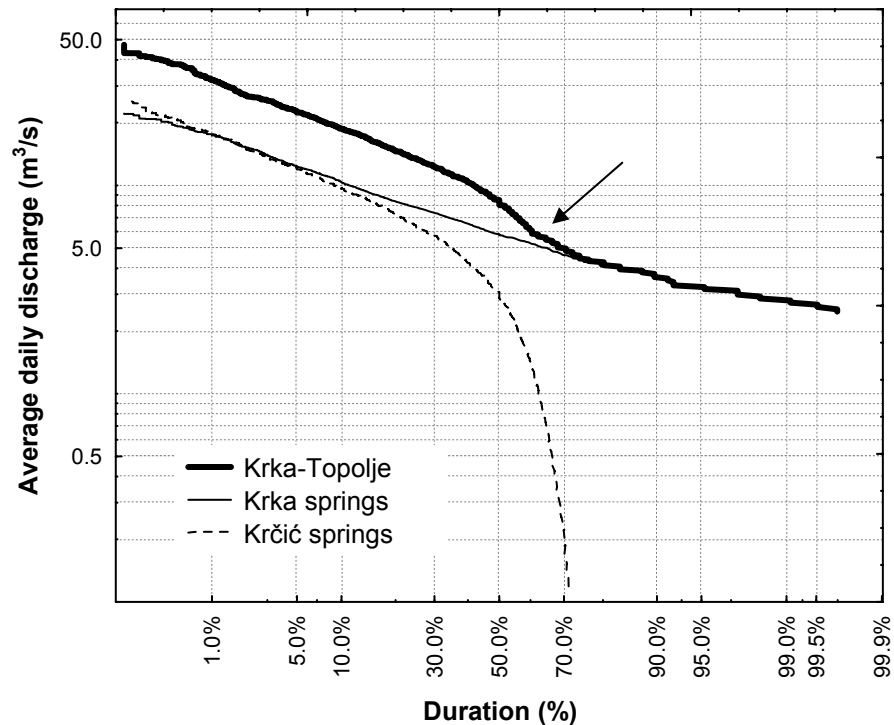


Fig. 5 Flow duration curves for the Krka-Topolje hydrometric station (Fig. 1, Station 1), Krka springs and Krčić springs, plotted on lognormal probability paper for the period 1982–1990.

The relationships between basic hydrological variables are analysed in this study by using monthly and annual groundwater recharge rates calculated by a mathematical model in order to: (a) estimate the hydrological catchment areas of the Krčić and Krka springs; (b) validate the hypotheses about the hydrological functioning elaborated in previous studies; and (c) investigate connections between the aquifers of the Main Krčić Spring and the adjacent spring zones. The results are presented in the following section.

GROUNDWATER RECHARGE ANALYSES

According to White (2002), the groundwater recharge of a karst aquifer can be divided into the following components: allogenic recharge through the swallets; diffuse infiltration entering the aquifer as infiltration through the soil and fractures and fissures; and internal runoff entering the aquifer quickly through sinkhole drains and overflow from perched aquifers. In this study, groundwater recharge is calculated by the mathematical model based on Palmer's (1965) soil-moisture balance method (Denić-Jukić, 2002). The groundwater recharge in this context represents the effective rainfall, i.e. the portion of total rainfall that instantaneously recharges the phreatic zone of an aquifer. This method implies that the initial precipitation left after evapotranspiration is taken up as soil moisture until the soil is saturated, after which the remaining water enters the aquifer quickly through sinkhole drains or fractures and fissures. This takes into account only the quick component of diffuse infiltration and internal runoff. The

parameters are the soil-moisture holding capacities of the upper (s_1) and lower (s_2) soil layers. From the upper layer, which implicitly includes land cover, the moisture is lost freely in the processes of evaporation and transpiration, while the moisture lost from the lower layer depends on the saturation of the upper layer. The potential moisture lost in the process of evapotranspiration is calculated by using the expression of Eagan (1967):

$$ET_p = C \cdot e_{\max} \cdot (100 - RH)^{1/2} ; e_{\max} = 6.1 \cdot e^{17.1T/(234.2+T)} \quad (1)$$

where ET_p is the potential evapotranspiration (mm), RH is the relative humidity (%) and T is the air temperature ($^{\circ}\text{C}$). The coefficient C depends on T : $C = 0.63$ for $T < 0^{\circ}\text{C}$; $C = 0.63 + 0.024T$ for $0 < T < 21^{\circ}\text{C}$; and $C = 1.13$ for $T > 21^{\circ}\text{C}$. The values of the model parameters are estimated in the frequency domain by the spectral method (Jukić & Denić-Jukić, 2004) and are presented in Table 5. Low values of parameter s_1 for the Krka and Krčić springs confirm that land cover has a limited influence on the groundwater recharge in karst (Bonacci, 1987). The values of parameter s_2 show that the influence of the lower soil layers is much more significant. The parameter s_2 for the Krčić springs ranges widely from 8 to 109 mm. The values for the Krka springs range from 23 to 69 mm and are 15 mm lower on average. The wide ranges of the values of parameter s_2 confirm that the soil-moisture holding capacities cannot be considered as a time-invariant characteristic of karst aquifers, especially if short time periods are analysed. However, the ranges and average values of the two model parameters obtained are in accordance with the results of previous model applications in the Dinaric karst (Denić-Jukić, 2002; Denić-Jukić & Jukić, 2003; Jukić & Denić-Jukić, 2004).

Table 5 The values of the parameters of the groundwater recharge model based on Palmer's (1965) soil-moisture balance method determined by the spectral method.

Hydrological year	Krka springs:		Krčić springs:	
	s_1 (mm)	s_2 (mm)	s_1 (mm)	s_2 (mm)
1982/83	5	31	0	28
1983/84	1	49	0	48
1984/85	9	23	0	109
1985/86	6	69	2	66
1986/87	1	32	0	8
1987/88	0	37	0	93
1988/89	8	36	6	40
1989/90	0	34	0	38
Average	4	39	1	54

s_1 : soil-moisture capacity of the upper soil layer; s_2 : soil-moisture capacity of the lower soil layer.

The monthly results of the application of the groundwater recharge model are demonstrated in Fig. 6. Figure 6(a) presents the input time series of monthly rainfall rates for the period of eight hydrological years, 1982/82–1989/90. Figure 6(b) presents the monthly groundwater recharge rates for the Krka springs that were obtained with Palmer's (1965) soil-moisture balance method for the values of parameters s_1 and s_2 in Table 5. The monthly effective infiltration coefficients in Fig. 6(c) were calculated as the ratio between the monthly groundwater recharge rates and the monthly rainfall

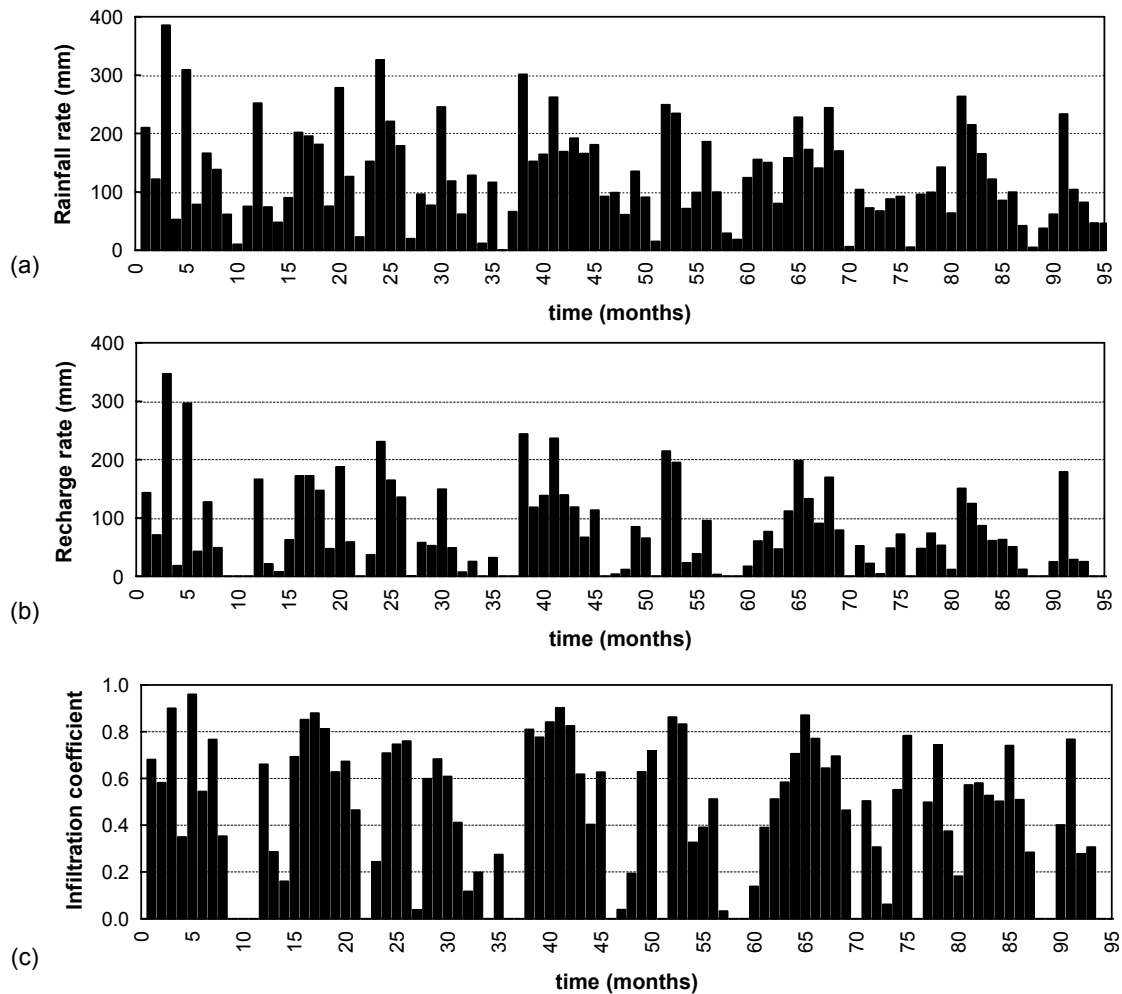


Fig. 6 (a) Monthly rainfall rates, (b) monthly groundwater recharge rates, and (c) effective infiltration coefficients calculated by the groundwater recharge model for the Krka springs for eight hydrological years (1982/83–1989/90).

rates. The distribution of the groundwater recharge rates is highly irregular. Rainy months without any groundwater recharge are followed by months having high values of effective infiltration coefficients: values range from 0 in July and August (complete monthly rainfall is spent for evapotranspiration) to >0.9 during rainy months in the cold seasons.

The polynomial regression between the monthly groundwater recharge rates and the average monthly discharges from the Krka springs is presented in Fig. 7. One can see a limit for monthly groundwater recharge rates between 150 and 200 mm. The monthly recharge above this limit does not contribute significantly to the Krka springs discharge. It can also be noted from Fig. 7 that the Krka springs have an important baseflow component of discharge that is practically independent of the monthly groundwater recharge. Unlike the Krka springs, there is no statistically significant correlation between the monthly groundwater recharge rates and the average monthly discharges from the Krčić springs, because of the considerable seasonal storage capacity of the aquifer and catchment (snow cover on the Dinara Mountains), which redistributes water during the year.

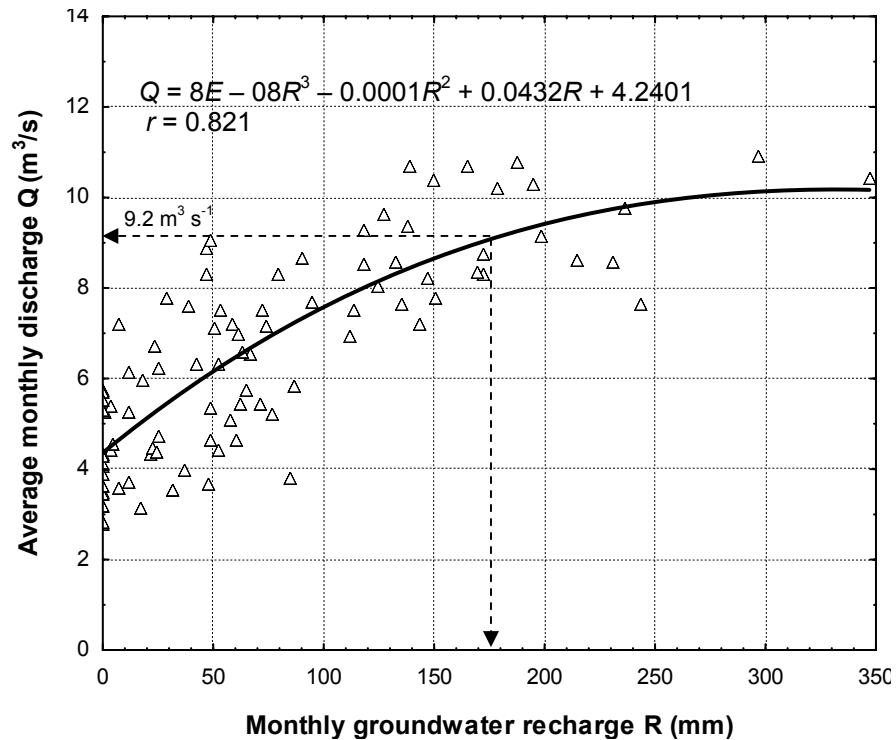


Fig. 7 Cubic-order polynomial regression between monthly groundwater recharges and average monthly discharges from the Krka springs.

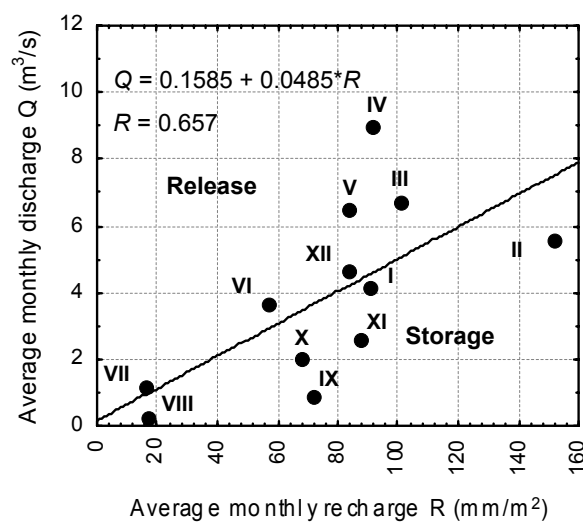


Fig. 8 Linear regression between the average monthly groundwater recharge for each month in the year (I: January ... XII: December) and the average monthly discharge of the Krčić springs.

The linear regression between average monthly recharges for each month in the year (I: January–XII: December) and average discharges from the Krčić springs for each month is presented in Fig. 8. Although the linear regression coefficient is not significant, the regression line can be considered as a delimiter for the estimation of the intensity of water storage (area below the line) and water release (area above the line)

in each month, or the seasonal storage capacity of the aquifer and catchment. It shows that water is usually stored between August and February (except December) and it is released between March and July. The largest amount of water is stored during September, when the aquifer is empty following the summer, and released during April, due to snowmelt.

The annual results of the application of the groundwater recharge model are presented in Table 6. The average annual effective infiltration coefficients, C_A , and the average annual hydrological catchment areas, A_A , are calculated by the following expressions:

$$C_A = \frac{R_A}{P_A}; A_A = \frac{V_A}{R_A} \quad (2)$$

where R_A is annual groundwater recharge (Table 6), P_A is the annual rainfall rate (Table 2) and V_A is total annual discharge (Table 2). If the average catchment area values for the entire period 1982–1990 in Table 6 are compared with those obtained by the regional formula and empirical expressions (Table 4), it can be seen that the groundwater recharge model produced results that are very similar to those of the empirical expression of Turc (1954).

Table 6 Annual results of the groundwater recharge model application to the Krčić and Krka springs. For the Krčić springs, the annual groundwater recharge values are given separately for the whole period and for the period when the springs are dry.

Period	Krčić springs: R_A (m m ⁻²)		C_A	A_A (km ²)	Krka springs:		
	Total	Dry period			R_A (m m ⁻²)	C_A	A_A (km ²)
1982/83	1.154	0.019	0.712	128.6	1.095	0.676	148.5
1983/84	1.100	0.030	0.632	107.2	1.087	0.624	216.6
1984/85	0.931	0.008	0.591	140.1	0.897	0.570	229.8
1985/86	1.229	0.043	0.645	154.5	1.191	0.625	194.5
1986/87	0.899	0.114	0.698	127.2	0.721	0.560	257.7
1987/88	0.904	0.003	0.488	141.9	1.068	0.577	239.7
1988/89	0.569	0.031	0.530	129.7	0.577	0.537	314.3
1989/90	0.519	0	0.486	138.2	0.532	0.498	360.3
1982–1990	7.305	0.249	0.603*	133.5*	7.168	0.591*	230.3*

R_A : groundwater recharge; C_A : effective infiltration coefficient; A_A : hydrological catchment area.

*Average values

The linear regressions between annual groundwater recharges and total annual discharges for the Krčić and Krka springs, with 95% confidence intervals for the regression lines, are presented in Fig. 9. Just as in Fig. 7, a component of the Krka springs discharges that is almost independent of the groundwater recharges is visible (Fig. 9(a)), although the linear regression coefficient is not statistically significant. This component confirms the existence of an important multi-annual storage in the Krka springs aquifer. The linear relationship between the groundwater recharge and total annual discharge values of the Krčić springs is evident (Fig. 9(b)). The regression line almost passes through the origin of the co-ordinate system, as does the regression line in Fig. 8, which means that the Krčić springs aquifer has insignificant outflows

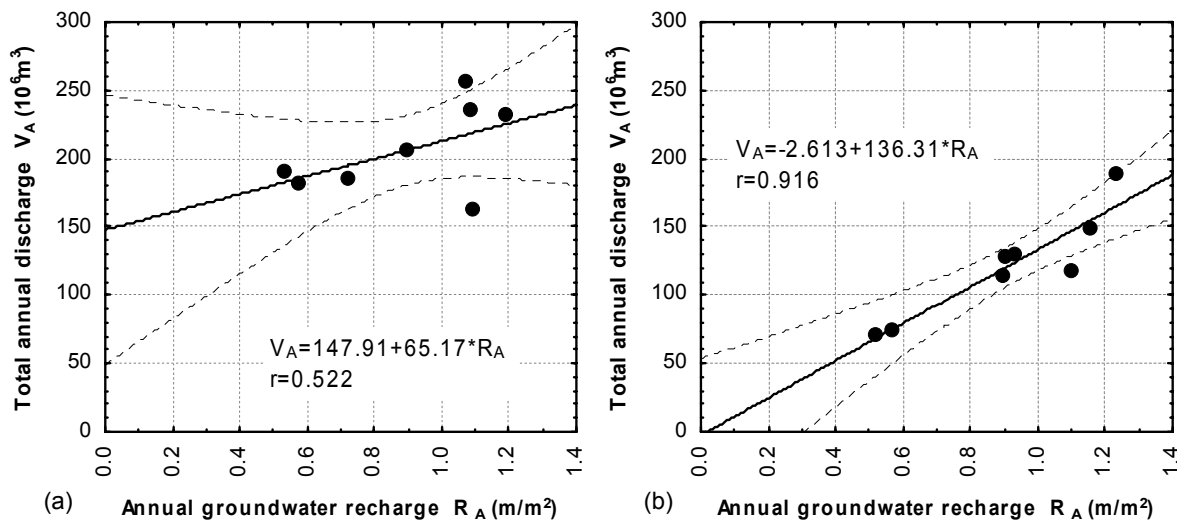


Fig. 9 Linear regressions between the annual groundwater recharge and the total annual discharge from (a) the Krka springs and (b) the Krčić springs, showing 95% confidence intervals for the regression lines.

toward adjacent aquifers. In Table 5, annual groundwater recharge is given separately for the period when the Krčić springs are dry. It can be noted that only 4% of the total groundwater recharge for the period 1982/83–1989/90 belongs to the dry period. The conclusion is that the storage capacity of the part of aquifer located below the level of the Main Krčić Spring is small, so the dynamic water storage is located mainly above the spring level. Thus the spring functions virtually as a descending karst spring, which confirms the results obtained by Jukić (2005a). Consequently, groundwater circulation from the Main Krčić Spring aquifer toward the Krka springs is nearly negligible from the standpoint of the total annual discharges from the Krčić and Krka springs; this is in agreement with the form of flow duration curves presented in Fig. 5.

The effective infiltration coefficients for the Krka springs range from 0.498 for the hydrological year 1989/90 to 0.676 for 1982/83, with a mean value of 0.591 (Table 6). The values for the Krčić springs range even more widely, but the mean value of 0.603 is very similar to that for the Krka springs. The hydrological catchment area for the Krka springs has the widest range of values, from as low as 148.5 to a high value of 360.3 km^2 . In Fig. 10(a), the hydrological catchment areas for the Krka springs are compared with the effective infiltration coefficients. The linear relationship is evident, which indicates that the hydrological catchment area is dependent upon soil moisture. Linear regression between the volumes of discharges greater than $9.2 m^3 s^{-1}$ and the hydrological catchment areas (Fig. 11) also shows that the Krka springs catchment is larger during dry hydrological years. This finding is in agreement with the results of polynomial regression between the monthly groundwater recharge rates and the average monthly discharges from the Krka springs presented in Fig. 7, and it confirms the existence of time-variant catchment boundaries, or outflows that occur during high water levels. In order to find out the possible directions of the outflows, the regressions between the hydrological catchment areas of the Krka springs and the total annual discharges from adjacent springs were tested. The strongest linear regression was found for the Cetina springs discharges (Fig. 10(b)).

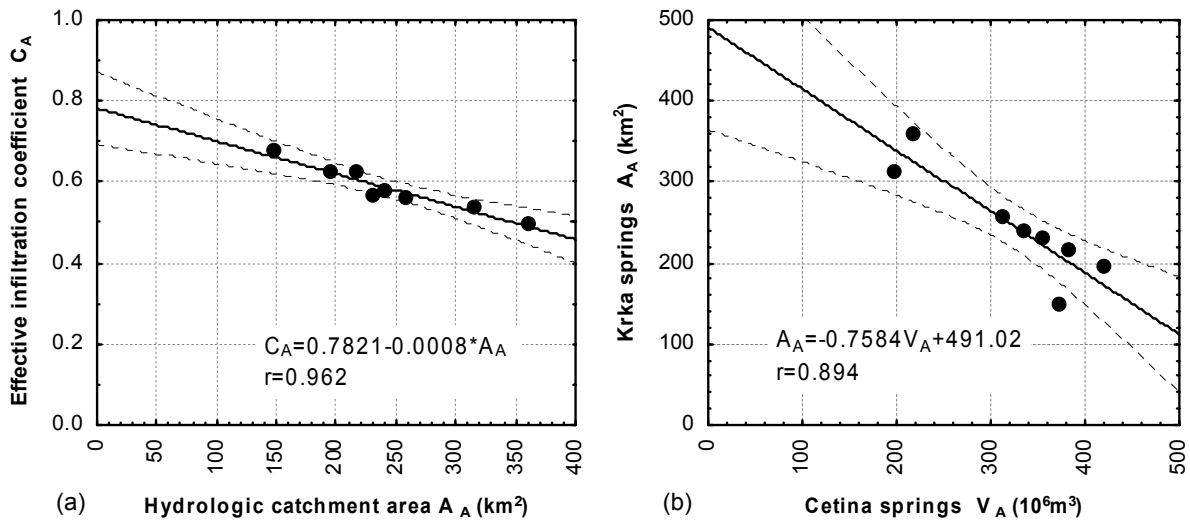


Fig. 10 (a) Linear regression between the hydrological catchment area and the effective infiltration coefficient for the Krka springs with 95% confidence interval for the regression line. (b) Linear regression between the total annual discharge of the Cetina springs and the hydrological catchment area of the Krka springs.

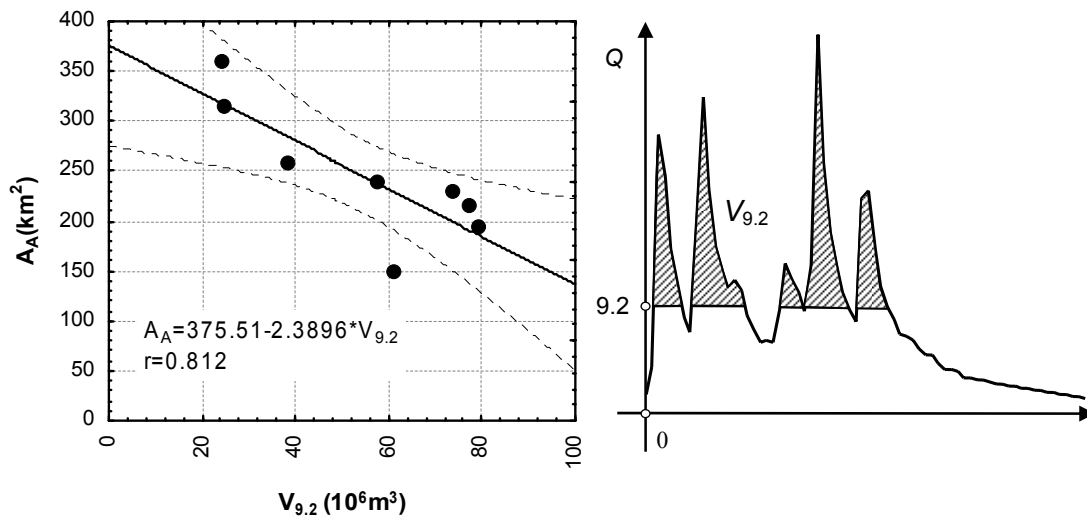


Fig. 11 Linear regression between the volume of discharges greater than $9.2 \text{ m}^3 \text{ s}^{-1}$ ($V_{9.2}$) and the hydrological catchment area for Krka springs (A_A) with 95% confidence interval for the regression line.

The results of the analyses depict complexity of the groundwater circulations between the spring zones of the Krčić, Krka and Cetina river catchments. The Krka springs hydrological catchment area is between 150 and 360 km^2 , depending on the groundwater levels. The topographic catchment area of the Krka springs, including the Main Krčić Spring topographic catchment, is only 145 km^2 (area $A_1 + A_2$ in Fig. 1), so approximately up to 240 km^2 is situated outside topographic catchment boundaries in the Pašića Polje area (area A_4 in Fig. 1). From this area, through the ponors, water comes to the small springs located near the right bank of the River Krčić, and also to the Krka springs, by deep percolation beneath the catchment area of the River Krčić, in

accordance with the hypothesis of Bonacci & Ljubenkov (2005), (Fig. 4). The Pašića Polje is the part of the Krka springs catchment that is also shared with the Cetina springs during high water levels. The average annual hydrological catchment area of the Main Krčić Spring is about 135 km². The topographic catchment area of this spring is only 35 km² (area A₁ in Fig. 1), which means that approximately 100 km² is situated outside. Hydrogeological parameters (directions of faults), considerable redistribution of water during the year due to snow cover (Fig. 8) and similar hydrological behaviour of the Main Krčić Spring and Cetina springs discharges (Jukić, 2005b) indicate that this additional area is located in the Dinara Mountains (area A₃ in Fig. 1). This part of the Main Krčić Spring catchment is located within the Cetina springs topographic catchment and is partly shared with the Cetina springs. Consequently, the hydrological connections between the Main Krčić Spring and the Cetina springs are stronger than those between the Main Krčić Spring and the Krka springs. The Main Krčić Spring functions as a descending karst spring. Subsurface flow from the Main Krčić Spring aquifer toward the Krka springs is practically negligible from the standpoint of the total annual discharges, so an intensive flow through highly karstified zone along the Krčić river bed (Fig. 3) exists only when the river is active.

CONCLUSION

In this study, classical hydrological and some more modern time and frequency domain methods were used for qualitative and quantitative analysis. Basic hydrological analyses were performed in order to explain and supplement the existing data and information necessary in the study. Groundwater recharge rates were calculated by a mathematical model based on Palmer's (1965) soil-moisture balance method. Although this method was not developed originally for karst areas, the results of several applications in the Dinaric karst (Denić-Jukić, 2002; Denić-Jukić & Jukić, 2003; Jukić & Denić-Jukić, 2004) have shown that it can be successfully applied in preliminary analyses. The values of the model parameters were estimated in the frequency domain by the spectral method. The model parameter values obtained are in accordance with the results of previous model applications in the Dinaric karst. Since the average catchment area values for the entire period 1982–1990 obtained by the groundwater recharge model are also in accordance with those obtained by the formula and empirical expressions commonly applied in the Dinaric karst, the applicability of the methodology to the analysed catchments is confirmed. The calculated monthly and annual groundwater recharge rates formed the basis for the estimation of hydrological catchment areas and also for the determination of quantitative relationships between the catchments in order to validate the existing hypotheses and contribute to factual information about hydrological behaviour of the three spring zones. Data were available for only eight hydrological years, so the simplest linear regression model was applied, in the first place, to discover the existing relationships between hydrological variables without presuming that the relationships are actually linear.

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